

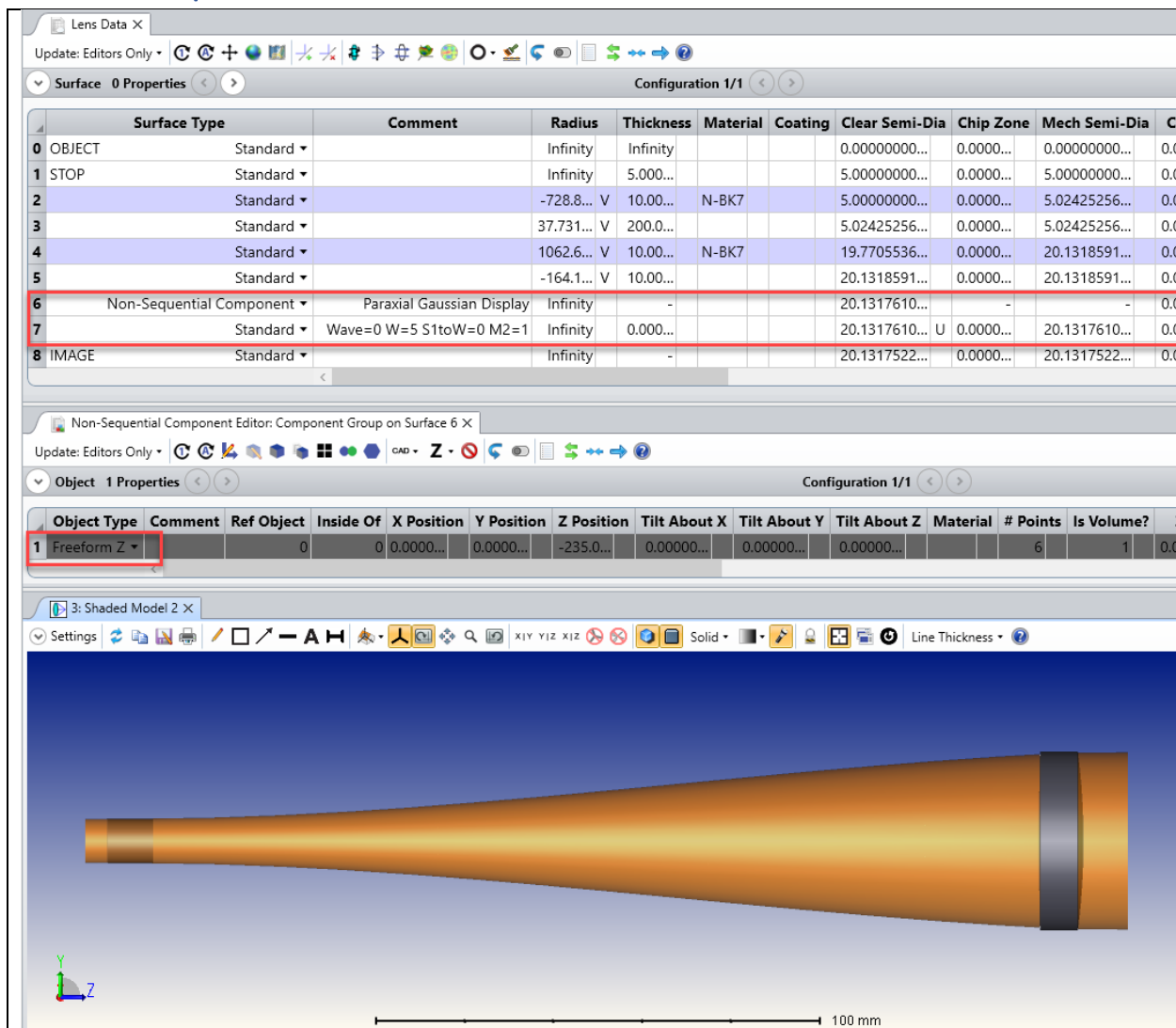
Title

API_Extension_CS_Sequential_Solid_Gaussian_Beam_Envelope

Author

Sandrine Auriol - ZEMAX

Brief Description



Overview

This Zemax User-Extension displays the envelope of a Gaussian Beam. The Gaussian Beam characteristics are calculated by the Paraxial Gaussian Beam calculator. The extension adds a Non-Sequential Component at the bottom of the sequential editor. A Freeform Z object is added to the Non-

Sequential Component Editor. It is formed with the different sizes of the beam calculated by the Paraxial Gaussian Beam

Plugin Type

API Extension

Language

CS

Updates

Date	Version	OpticStudio version	Comment
24/03/2020	1.0	20.1.1	Creation

Instructions

1. Installation

To install the Zemax User Plugin, copy the files called GaussianBeamDisplay.pdb, GaussianBeamDisplay.exe.config and GaussianBeamDisplay.exe under {Zemax}\ZOS-API\Extensions.

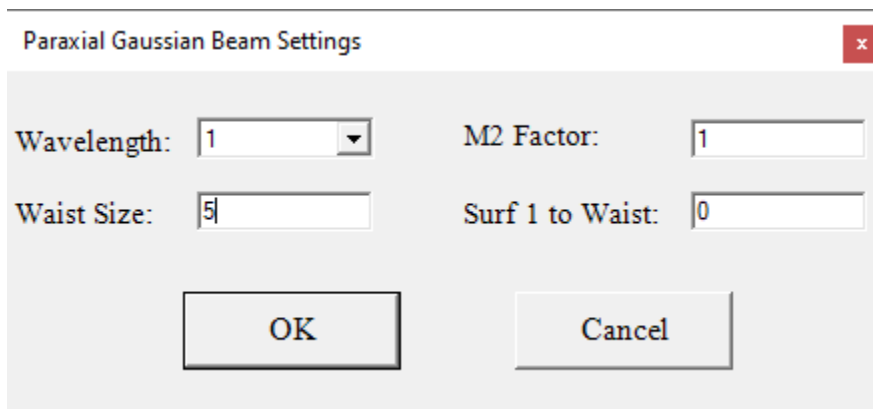
OpticStudio will have to be restarted to be able to view the newly installed Zemax User Plugin.

2. How to use

To use the Zemax User plugin, open a sequential file.

Go to Programming Tab...User Analyses and select the Zemax Plugin.

First select the settings:



Paraxial Gaussian Beam Settings

Wavelength:	1	M2 Factor:	1
Waist Size:	5	Surf 1 to Waist:	0

OK Cancel

Once you press OK, the software will calculate the size of the Paraxial Gaussian Beam at every surface. It will then create a Freeform Z object that models the envelope of the Paraxial Gaussian Beam.

Code Exchange

Zemax

Lens Data X

Update: Editors Only

Surface 0 Properties

Configuration 1/1

	Surface Type	Comment	Radius	Thickness	Material	Coating	Clear Semi-Dia	Chip Zone	Mech Semi-Dia	Coat
0	OBJECT	Standard	Infinity	Infinity			0.00000000...	0.0000...	0.00000000...	0.0...
1	STOP	Standard	Infinity	5.000...			5.00000000...	0.0000...	5.00000000...	0.0...
2	Standard		-728.8...	V 10.00...	N-BK7		5.00000000...	0.0000...	5.02425256...	0.0...
3	Standard		37.731...	V 200.0...			5.02425256...	0.0000...	5.02425256...	0.0...
4	Standard		1062.6...	V 10.00...	N-BK7		19.7705536...	0.0000...	20.1318591...	0.0...
5	Standard		-164.1...	V 10.00...			20.1318591...	0.0000...	20.1318591...	0.0...
6	Non-Sequential Component	Paraxial Gaussian Display	Infinity	-			20.1317610...	-	-	0.0...
7	Standard	Wave=0 W=5 S1toW=0 M2=1	Infinity	0.000...			20.1317610... U	0.0000...	20.1317610...	0.0...
8	IMAGE	Standard	Infinity	-			20.1317522...	0.0000...	20.1317522...	0.0...

Non-Sequential Component Editor: Component Group on Surface 6 X

Update: Editors Only

Object 1 Properties

Configuration 1/1

	Object Type	Comment	Ref Object	Inside Of	X Position	Y Position	Z Position	Tilt About X	Tilt About Y	Tilt About Z	Material	# Points	Is Volume?	Z1
1	Freeform Z		0	0	0.0000...	0.0000...	-235.0...	0.00000...	0.00000...	0.00000...		6	1	0.0...

3: Shaded Model 2 X

Settings

Line Thickness

